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1.73 Norepinephrine

1.73.1 Preparation and Administration

Norepinephrine—Hypotension

Adult
- 2 - 35 mcg/min IV infusion
- Alternately 0.05 – 2 mcg/kg/min IBW IV infusion

Pediatric
- 0.05 – 2 mcg/kg/min IBW IV infusion

Norepinephrine—Hypotension related to sepsis

Adult
- 5-70 mcg/min infusion

Pediatric
- No PHI Standing Orders or Indications

1.73.2 Norepinephrine—Drug Information

Pharmacologic Category:	• Sympathomimetic
Contraindications:	• Hypersensitivity
Warnings / Precautions:	• Tachydysrhythmia
Adverse Reactions:	• Tachycardia, hypertension, tissue sloughing with extravasation
Pregnancy Risk Factors:	
ISMP High Alert:	• Yes
Onset / Peak / Duration:	• 1-3 min / 5 min / 5-10 min
Dose Scalar:	• Ideal Body Weight, Fixed
Mandatory Medication:	• Yes
Trade Name:	• Levophed

1.73.3 Norepinephrine—Advisories

- Must be infused through a large stable vein to avoid tissue necrosis
- If tissue sloughing occurs, the antidote is phentolamine 5 to 10 mg diluted in 10 mL 0.9% sodium chloride injected into extravasation area as soon as possible.
- May be combined with vasopressin and epinephrine as per Sepsis Resuscitation.

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1.73 Norepinephrine

1.73.1 Preparation and Administration

Norepinephrine—Hypotension

Adult
- 2 - 35 mcg/min IV infusion
- Alternately 0.05 – 2 mcg/kg/min IBW IV infusion

Pediatric
- 0.05 – 2 mcg/kg/min IBW IV infusion

Norepinephrine—Hypotension related to sepsis

Adult
- 5-70 mcg/min infusion

Pediatric
- No PHI Standing Orders or Indications

1.73.2 Norepinephrine—Drug Information

Pharmacologic Category:	• Sympathomimetic
Contraindications:	• Hypersensitivity
Warnings / Precautions:	• Tachydysrhythmia
Adverse Reactions:	• Tachycardia, hypertension, tissue sloughing with extravasation
Pregnancy Risk Factors:	
ISMP High Alert:	• Yes
Onset / Peak / Duration:	• 1-3 min / 5 min / 5-10 min
Dose Scalar:	• Ideal Body Weight, Fixed
Mandatory Medication:	• Yes
Trade Name:	• Levophed

1.73.3 Norepinephrine—Advisories

- Must be infused through a large stable vein to avoid tissue necrosis
- If tissue sloughing occurs, the antidote is phentolamine 5 to 10 mg diluted in 10 mL 0.9% sodium chloride injected into extravasation area as soon as possible.
- May be combined with vasopressin and epinephrine as per Sepsis Resuscitation.

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1.73.4 Norepinephrine mcg/min Infusion Chart

Norepinephrine mcg/min	4 mg/250 mL (16 mcg/mL)	8 mg/250 mL (32 mcg/mL)	16 mg/250 mL (64 mcg/mL)
2	7.5	3.8	1.9
3	11	5.6	2.8
4	15	7.5	3.8
5	19	9.4	4.7
6	23	11	5.6
7	26	13	6.6
8	30	15	7.5
9	34	17	8.4
10	38	19	9.4
11	41	21	10
12	45	23	11
13	49	24	12
14	53	26	13
15	56	28	14
16	60	30	15
17	64	32	16
18	68	34	17
19	71	36	18
20	75	38	19
21	79	39	20
22	83	41	21
23	86	43	22
24	90	45	23
25	94	47	23
26	98	49	24
27	101	51	25
28	105	53	26
29	109	54	27
30	113	56	28
31	116	58	29
32	120	60	30
33	124	62	31
34	128	64	32
35	131	66	33
	mL/hr		

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1.78.5 Phenylephrine Infusion Chart

Phenylephrine – Pediatric – 100 mcg/mL

Phenylephrine	100	mcg/mL							
mcg/kg/min	Weight In KG								
	5	10	15	20	25	30	35	40	45
0.1	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7
0.5	1.5	3	5	6	8	9	11	12	14
1	3.0	6	9	12	15	18	21	24	27
1.5	4.5	9	14	18	23	27	32	36	41
2	6.0	12	18	24	30	36	42	48	54
2.5	7.5	15	23	30	38	45	53	60	68
3	9.0	18	27	36	45	54	63	72	81
	mL/hr								

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1.78 Phenylephrine

1.78.1 Preparation and Administration

Phenylephrine—Hypotension—Infusion

Adult

- 100–180 mcg/min infusion
- Alternately 0.1–3 mcg/kg/min infusion
- Titrate to target BP

Ped

- 0.1–3 mcg/kg/min infusion
- Titrate to target BP

Phenylephrine—Hypotension—Push Pressor

Adult

- 50–200 mcg IV
- 100 mcg/mL concentration
- Repeat every 2–5 min PRN
- Administer up to 3 doses total

Ped

- 5–20 mcg/kg IV
- Max single dose 200 mcg
- 100 mcg/mL concentration
- Repeat every 2–5 min PRN
- Administer up to 3 doses total

1.78.2 Phenylephrine—Drug Information

Pharmacologic Category:	• Vasopressor
Contraindications:	• Ventricular Tachycardia
Warnings / Precautions:	• Septic shock, hyperthyroidism, bradycardia, partial heart block, myocardial disease
Adverse Reactions:	• Bradycardia, pulmonary edema, arrhythmias
Pregnancy Risk Factors:	
ISMP High Alert:	• Yes
Onset / Peak / Duration:	• Immediate/15–20 min/rapid offset
Dose Scalar:	• Fixed, Ideal Body Weight
Mandatory Medication:	• No
Tall Man Lettering:	• NA
Trade Name:	• Neo-synephrine

1.78.3 Phenylephrine—Advisories

- Maintenance infusions up to 300 mcg/min in adults and 5 mcg/kg/min in pediatrics for Tet spells may be required, upon consultation with online medical direction or managing physician.
- Recommended infusion mix 10 mg/100 mL (100 mcg/mL).
- Phenylephrine is pure alpha agent and potent vasoconstrictor suitable for hypotension secondary to anesthetic administration and vasodilatory shock.

IV push pressors should be used as a temporary treatment and limited to a maximum of three doses.

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1.68.4 Nicardipine Infusion

Nicardipine Infusion	0.1 mg/mL
Dose mg/hr	**Infusion Rate mL/hr**
3	30
5	50
7.5	75
10	100
12.5	125
15	150
Recommended Mix • 25 mg/250 mL—NS or D5W	

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1.53 Levetiracetam

1.53.1 Preparation and Administration

Levetiracetam—Seizures, Status Epilepticus

Adult

- 4 gm IV
- Administer over 15 min

Pediatric

- 60 mg/kg IV
- Administer over 15 min
- Max single dose 4 gm

1.53.2 Levetiracetam—Drug Information

Pharmacologic Category:	• Anticonvulsant
Contraindications:	• Hypersensitivity
Warnings / Precautions:	• CNS depression, renal impairment
Adverse Reactions:	• Hypertension, CNS and behavioral changes
Pregnancy Risk Factors:	
Onset / Peak / Duration:	• Immediate
Dose Scalar:	• Adult—Fixed Dose, Pediatric—Ideal Body Weight
Trade Name:	• Keppra

1.53.3 Levetiracetam—Advisories

Loading doses are not reduced in renal patients. Subsequent doses provided by direct medical control may need to be reduced based on glomerular filtration rate (GFR) and the type of dialysis therapy being used.

Usual Mix

- 2 gm/100 mL (20 mg/mL)
- 1 gm/50 mL (20 mg/mL)
- 1 gm/100 mL (10 mg/mL)
- 500 mg/50 mL (10 mg/mL)

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1.78.4 Phenylephrine Push Pressor Administration

IV push pressor phenylephrine should be used as a temporary treatment while preparing an infusion. Administration should be limited to 3 total doses.

Phenylephrine—Hypotension—Push Pressor

Adult

- 50–200 mcg IV
- 100 mcg/mL concentration
- Repeat every 2–5 min PRN
- Administer up to 3 doses total

Ped

- 5–20 mcg/kg IV
- Max single dose 200 mcg
- 100 mcg/mL concentration
- Repeat every 2–5 min PRN
- Administer up to 3 doses total

Phenylephrine Push Pressor Preparation

IV push pressor phenylephrine must be prepared and administered in the standard 100 mcg/mL concentration. Failure to do so may result in medication errors and patient harm.

Bag Dilution

1. Add 10 mg (1 ml via) phenylephrine to a 100 mL bag of NS—final concentration 100 mcg/mL
2. Label the bag
3. Using an empty syringe, draw the push dose phenylephrine to be administered from the bag
4. Label the syringe

The 100 mcg/mL phenylephrine IV bag is also an acceptable infusion mix and may be used for a follow-on infusion

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1.7 Amiodarone

1.7.1 Preparation and Administration

Amiodarone—V Fib/Pulseless V Tach

Adult
- 300 mg IV push
- Repeat 150 mg IV push PRN 1 time

Pediatric
- 5 mg/kg IV push
- Maximum single dose 300 mg
- Repeat 5 mg/kg PRN up to 2 times
- Maximum total dose 15 mg/kg

Amiodarone—Perfusing Tachycardias

Adult
- 150 mg mixed in 100 mL IV over 10 minutes
- Repeat 150 mg every 10 min PRN

Amiodarone—Perfusing Tachycardias

Pediatric
- 5 mg/kg IV over 20 – 60 min
- Maximum single dose 300 mg,
- Repeat 5 mg/kg IV over 20–60 min PRN up to 2 times

Amiodarone—Maintenance Infusion

Adult
- 1 mg/min for 6 hours, then
- 0.5 mg/min for 18 hours

Amiodarone—Maintenance Infusion

Pediatric
- 5–15 mcg/kg/minute
-

Amiodarone —Maintenance infusion

Neonatal
- 5–15 mcg/kg/minute

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1.37 Fentanyl

1.37.1 Dosing and Administration

Fentanyl—Analgesia

Adult
- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN

Pediatric
- Refer to adult dosing

Fentanyl—Post Intubation Pain and Agitation

Adult
- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN
- Titrate to sedation goal

Alternatively
- 1-2 mcg/kg/hr via infusion pump

Pediatric
- Refer to adult dosing

Fentanyl—Analgesia

Obstetrical
- 0.5–1 mcg/kg IV IBW over 1 min, IM, NAS
- Repeat every hour PRN

For pain related to headache or labor contractions

Do not administer opioid analgesics if delivery is possible within 1 hour of administration

Fentanyl—Analgesia

Neonatal
- 0.5 -1 mcg/kg IV over 1-2 min, IM, NAS
- Repeat every hour PRN

May use for analgesia or post intubation sedation

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1.64 Milrinone

1.64.1 Preparation and Administration

Milrinone—CHF

Adult

- 50 mcg/kg IV over 10 min loading dose
- Follow with continuous infusion at 0.25–0.75 mcg/kg/min
- Titrate to hemodynamic and clinical response

Milrinone—CHF, left sided valvular disease, pulmonary hypertension

Pediatric

- 0.25–0.75 mcg/kg/min IV

Milrinone—CHF, acute uncompensated heart failure

Neonatal

- 0.25–0.75 mcg/kg/min IV
- No loading dose

1.64.2 Milrinone—Drug Information

Pharmacologic Category:	• Inotropic, vasodilator
Contraindications:	• Severe aortic or pulmonic valvular disease
Warnings / Precautions:	• Elderly, pregnancy, and lactation
Adverse Reactions:	• Hypotension, SVT, thrombocytopenia, hypokalemia, ventricular arrhythmias
Pregnancy Risk Factors:	
ISMP High Alert:	• Yes
Onset / Peak / Duration:	• 5 min / 15 min / 4 hours
Dose Scalar:	• Ideal Body Weight
Mandatory Medication:	• No
Trade Name:	• Primacor

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1.73 Norepinephrine

1.73.1 Preparation and Administration

Norepinephrine—Hypotension

Adult

- 2 - 35 mcg/min IV infusion
- Alternately 0.05 – 2 mcg/kg/min IBW IV infusion

Pediatric

- 0.05 – 2 mcg/kg/min IBW IV infusion

Norepinephrine—Hypotension related to sepsis

Adult

- 5-70 mcg/min infusion

Pediatric

- No PHI Standing Orders or Indications

1.73.2 Norepinephrine—Drug Information

Pharmacologic Category:	• Sympathomimetic
Contraindications:	• Hypersensitivity
Warnings / Precautions:	• Tachydysrhythmia
Adverse Reactions:	• Tachycardia, hypertension, tissue sloughing with extravasation
Pregnancy Risk Factors:	
ISMP High Alert:	• Yes
Onset / Peak / Duration:	• 1-3 min / 5 min / 5-10 min
Dose Scalar:	• Ideal Body Weight, Fixed
Mandatory Medication:	• Yes
Trade Name:	• Levophed

1.73.3 Norepinephrine—Advisories

- Must be infused through a large stable vein to avoid tissue necrosis
- If tissue sloughing occurs, the antidote is phentolamine 5 to 10 mg diluted in 10 mL 0.9% sodium chloride injected into extravasation area as soon as possible.
- May be combined with vasopressin and epinephrine as per Sepsis Resuscitation.

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1.73.5 Norepinephrine mcg/kg/min Adult Infusion

Norepinephrine	4 mg/250 mL (16 mcg/mL)									
mcg/kg/min	Ideal Body Weight In kg									
	55	60	65	70	75	80	85	90	95	100
0.05	10	11	12	13	14	15	16	17	18	19
0.1	21	23	24	26	28	30	32	34	36	38
0.2	41	45	49	53	56	60	64	68	71	75
0.3	62	68	73	79	84	90	96	101	107	113
0.4	83	90	98	105	113	120	128	135	143	150
0.5	103	113	122	131	141	150	159	169	178	188
0.6	124	135	146	158	169	180	191	203	214	225
0.7	144	158	171	184	197	210	223	236	249	263
0.8	165	180	195	210	225	240	255	270	285	300
0.9	186	203	219	236	253	270	287	304	321	338
1	206	225	244	263	281	300	319	338	356	375
1.5	309	338	366	394	422	450	478	506	534	563
2	413	450	488	525	563	600	638	675	713	750
mL/hr										

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1.35 Esmolol

1.35.1 Preparation and Administration

Esmolol—Aortic Dissection, Tachyarrhythmia

Adult

- Initial load: 500 mcg/kg IV over 1 min
- Titrate 50 mcg/kg/min every 5 min to target HR or BP
- Repeat the loading dose of 500 mcg/kg before every titration PRN until near target HR
- Max infusion dose 300 mcg/kg/min

Pediatric

- No PHI Standing Orders or Indications

1.35.2 Esmolol—Drug Information

Pharmacological Category:	• Antiarrhythmic Agent, Class II; Antihypertensive; Beta-Blocker, Beta-1 Selective
Contraindications:	• Severe sinus bradycardia; heart block greater than first degree, sick sinus syndrome • Cardiogenic shock; uncompensated heart failure; pulmonary hypertension
Warnings / Precautions:	• In general, patients with bronchospastic disease should not receive beta-blockers; however, esmolol, with B_1 selectivity, has been used cautiously with close monitoring
Adverse Reactions:	• Hypotension, Bradycardia, Asystole
Pregnancy Risk Factors:	
Onset / Duration:	• 2–10 min/10–30 min
Dose Scalar:	• Ideal Body Weight
Trade Name:	• Brevibloc

1.35.3 Esmolol—Advisories

- In the event of bronchospasm, stop the infusion immediately and consult medical control.
- Esmolol may decrease fetal heart rate

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1.12 Calcium Chloride

1.12.1 Preparation and Administration

Calcium Chloride—Hyperkalemia, Crush Syndrome

Adult
- 1 gm IV over 2–10 min
- Repeat every 10 minutes PRN up to 3 times

Pediatric
- 20 mg/kg IV over 2–10 min
- Repeat in 10 minutes PRN
- Maximum single dose 1 gram

Faster rate of IV administration may be indicated in cardiac arrest, cardiovascular instability and crush injuries
Flush the IV line after administration

Calcium Chloride—Blood Transfusion

Adult
- 1 gm IV over 2–10 min
- Repeat each time 2 units of blood products are given

Pediatric
- 20 mg/kg IV over 2–10 min
- Repeat each time 20 mL/kg of blood products are given
- Maximum single dose 1 gram

Give if there is no improvement after blood administration
Flush the IV line after administration

Calcium Chloride—Magnesium Toxicity

Obstetric
- 0.5–1 gm IV over 3 min
- May repeat x 1 if symptoms do not resolve

1.12.2 Calcium Chloride—Drug Information

Pharmacologic Category:	• Electrolyte
Contraindications:	• Hypersensitivity, hypercalcemia, digitalis toxicity
Warnings / Precautions:	• Digitalis
Adverse Reactions:	• Brady-asystolic arrest, hypotension, tissue necrosis with extravasation
Pregnancy Risk Factors:	
ISMP High Alert:	• No
Onset / Peak / Duration:	• Seconds/3-5 min/30 min
Dose Scalar:	• Fixed Dose, Ideal Body Weight
Mandatory Medication:	• No
Trade Name:	• NA

1.12.3 Calcium Chloride—Advisories

- Mixing calcium with other medications may cause a precipitate to form in the IV line
- Administer to neonates with on-line medical direction
- Rapid administration may cause bradycardia and hypotension
- Monitor patients taking digoxin for acute toxicity which can develop with calcium infusion

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1.73.5 Norepinephrine mcg/kg/min Adult Infusion

Norepinephrine	4 mg/250 mL (16 mcg/mL)									
mcg/kg/min	Ideal Body Weight In kg									
	55	60	65	70	75	80	85	90	95	100
0.05	10	11	12	13	14	15	16	17	18	19
0.1	21	23	24	26	28	30	32	34	36	38
0.2	41	45	49	53	56	60	64	68	71	75
0.3	62	68	73	79	84	90	96	101	107	113
0.4	83	90	98	105	113	120	128	135	143	150
0.5	103	113	122	131	141	150	159	169	178	188
0.6	124	135	146	158	169	180	191	203	214	225
0.7	144	158	171	184	197	210	223	236	249	263
0.8	165	180	195	210	225	240	255	270	285	300
0.9	186	203	219	236	253	270	287	304	321	338
1	206	225	244	263	281	300	319	338	356	375
1.5	309	338	366	394	422	450	478	506	534	563
2	413	450	488	525	563	600	638	675	713	750
mL/hr										

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1.86 Rocuronium

1.86.1 Preparation and Administration

Rocuronium—Intubation, RSI

Adult
- 1.5 mg/kg IBW IV push over 2–3 sec
- Repeat PRN

Pediatric
- Refer to adult dosing

Rocuronium—Post Intubation Neuromuscular Blockade

Adult
- 1 mg/kg IBW IV over 10-15 sec
- Repeat PRN

Pediatric
- Refer to adult dosing

Rocuronium—Dangerous Agitation

Adult
- 1.5 mg/kg IBW IV, IO or
- 2 mg/kg IM (onset 5 min)

Pediatric
- No PHI Standing Order Or Indication

1.86.2 Rocuronium—Drug Information

Pharmacologic Category:	• Non-depolarizing neuromuscular blocker
Contraindications:	• Hypersensitivity
Warnings / Precautions:	• Theophylline increases risk of dysrhythmias
Adverse Reactions:	• Prolonged blockade, bronchospasm, vomiting
Pregnancy Risk Factors:	
Onset / Peak / Duration:	• IV or IO - 1 min / 1-5 min / 20-30 min - Dose dependent
Dose Scalar:	• Ideal Body Weight
Trade Name:	• Zemuron

1.86.3 NMB—Advisories

Neuromuscular blocking agents causes paralysis and respiratory arrest. Capabilities for advanced airway management, mechanical ventilation and oxygen administration must be readily available prior to NMB administration.

Neuromuscular blocking agents have no effect on consciousness, pain threshold, or cerebration. NMB administration must be accompanied by adequate analgesia and sedation**.**

Pain and sedation scales such as CPOT, BPS, SAS and RASS, measure behaviors that a paralyzed patient cannot display. These scores can not be used to assess and direct analgesia and anxiolysis in patients that have received NMBs.

- Segregate and properly label neuromuscular blocking agents to avoid confusion with other injectable solutions.
- Neuromuscular blocking agents have no effect on neuronal seizure activity.

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1.68 Nicardipine

1.68.1 Preparation and Administration

Nicardipine—AIS, ICH, AoD BP Management

Adult

- 5 mg/hr IV infusion
- Increase by 2.5 mg/hr every 5–15 minutes PRN
- Titrate to BP target
- Max 15 mg/hr

The dose may be reduced to 3 mg/hr after targeted response is achieved

Ped

- No PHI Standing Orders or Indications

1.68.2 Nicardipine—Drug Information

Pharmacologic Category:	• Antihypertensive, calcium channel blocker
Contraindications:	• Sick sinus syndrome, 2nd and 3rd degree heart blocks, advanced aortic stenosis
Warnings / Precautions:	• Breastfeeding, CHF, hypotension, hepatic injury, renal disease, Increases effect of digoxin, theophylline, alcohol, nitrates, quinidine, propanolol and NMB
Adverse Reactions:	• Bradycardia, hypotension; congestive heart failure; nausea, vomiting, Steven Johnson syndrome, acute renal failure, dysrhythmia
Pregnancy Risk Factors:	
ISMP High Alert:	• Yes
Onset / Peak / Duration:	• Immediate/immediate/4–6 hours
Dose Scalar:	• Dose
Mandatory Medication:	• No
Tall Man Lettering:	• NA
Trade Name:	• Cardene

1.68.3 Nicardipine—Advisories

- Cimetidine increases nicardipine levels

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1.63 Midazolam

1.63.1 Preparation and Administration

Midazolam—Anxiolysis

Adult
- 0.5–5 mg IV over 1-2 min, IM, NAS
- Repeat every 5 min PRN

Pediatric
- 0.05–0.1 mg/kg IBW IV over 1-2 min, IM, NAS
- Maximum single dose 5 mg
- Repeat every 2 min PRN

Midazolam—At Risk for Dangerous Agitation

Adult
- 5 mg IV, IM, or IO
- Repeat in 5 min if needed

Droperidol is first line treatment for At Risk Dangerous Agitation. Midazolam co-therapy may be considered. Consider a 50% dose reduction in elderly patients or those with significant comorbidities to reduce the risk of respiratory depression and prolonged sedation.

Midazolam—At Risk for Dangerous Agitation

Pediatric
- No PHI Standing Orders or Indications

Midazolam—RSI

Adult
- 0.05–0.2 mg/kg IV over 1-2 min

Pediatric
- Refer to adult dosing

Midazolam—Post Intubation Agitation

Adult
- 0.025 to 0.05 mg/kg IV
- Repeat every 10 min PRN to achieve the sedation goal
- Repeat PRN for breakthrough agitation

To maintain goal
- Repeat bolus every 20–60 min, OR
- 0.05 to 0.2 mg/kg/hr continuous infusion

Pediatric
- Refer to adult dosing

Midazolam is the preferred sedative agent if there is a likelihood of seizure activity masked by NMB.

Midazolam—Seizure

Adult
- 0.1 mg/kg IV
- Max single dose 5 mg

Alternately
- 0.2 mg/kg NAS, IM
- Max single dose 10 mg

Pediatric
- Refer to adult dosing

Preferred benzodiazapine if IM or NAS administration is indicated.
Repeat every 2–3 minutes until the seizure is controlled.

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